Regis University
College for Professional Studies
School of Computer & Information Sciences

SYLLABUS

COURSE NUMBER: MT 415

COURSE TITLE: Linear Algebra

Course Description

Studies vector spaces, linear transformations, matrices, determinants, systems of equations, Eigen values and characteristic matrices.

Course Prerequisites

MT 360A Calculus I

Course Outcomes

- Demonstrate the ability to communicate the concepts of linear algebra, so that you may provide the public with information and expert judgment as part of the basis for social and political decisions.
- Demonstrate an awareness of the socially beneficial uses of mathematics, as well as harmful uses of mathematics.
- Describe systems of linear equations and implement their solutions.
- Explain the concepts of vectors, vector spaces, and subspaces.
- Demonstrate the concepts of matrix operations, partitioned matrices, matrix factorization, symmetric matrices, and quadratics form.
- Demonstrate knowledge of determinants, their properties, and rank of matrices.
- Demonstrate understanding of transformations, coordinate systems and graphical interpretation of coordinates.
- Describe Eigen vectors, Eigen values, and their applications.
- Explain the concepts of orthogonal vectors, sets, and projections.

Course Materials and Resources, including Technology Tools

Required:

Lay, D. C. (2012). *Linear algebra and its applications* (4th ed.). Boston, MA: Addison-Wesley (Pearson). ISBN13: 978-0-321-83614-4 or 0-321-83614-6 hardcover (text with MyMathLab).

Graphing calculator with matrix capability. Calculators with the requisite capabilities include the HP48S, HP48SX, or the TI-85. Each of these may require some add-on programs to perform

special operations on vectors and matrices. Generally, no instruction will be provided for the calculators. The student may use other programs such as MathCAD® or Matlab® to help in doing homework, but these programs will not be available for use on an exam.

Supplemental:

Lipschutz, Seymour, (1989). *3000 Solved Problems in Linear Algebra*, New York, NY: McGraw Hill Professional Publishing; ISBN10: 0-07-038023-6 or ISBN13: 9780070380233.

Method of Instruction

This course will use lab exercises, facilitated discussions, and lectures.

Related Standards

Related Standards	
	This program enables students to achieve the following program outcomes by the time of graduation:
	An ability to apply knowledge of computing and mathematics appropriate to the discipline
	An ability to analyze a problem, and identify and define the computing requirements appropriate to its solution
	An ability to communicate effectively with a range of audiences
	An ability to analyze the local and global impact of computing on individuals, organizations, and society
Regis Nine Educational Outcomes	
	Content Knowledge in Discipline
	Diverse Cultures, Perspectives, and Belief Systems
	Arts, Sciences, and Humanities
	Critical Thinking
	Communication
	Technology
	Ethical and Social Responsibility
	Leadership in Service to Others
	Lifelong Learning
Jesuit Themes - These are themes that are embedded as part of our interactions and instruction at Regis, but are not written in as activities	
	Magis, Finding God in All Things, Unity of Hearts and Mind, Contemplatives in Action, Care of the Whole Person, Men and Women for Others

First Week Assignment

Regis University has implemented a "Writing across the Curriculum Program." Because of the nature of this course and the amount of material to be covered, you will be required to submit

only one written paper. That paper will be due on the first day of class. The first assignment must be turned in at the first class meeting or when you first log into your on-line class to receive a grade.

The first assignment will consist of writing a 2-page, type written paper following the outline below:

1. A brief biography to include what you do for a living
2. Your experience in mathematics to include both high school and college
3. What do you expect to get out of this course (other than a grade)

Course Overview Grid

Week	Learning Topics	Activities	Reading Assignments	Assignments
1	Solutions of Linear Equations	Lab exercises, lectures, and facilitated discussions	Chapter 1: Sections 1 to 6	First Week Writing Assignment Solve the following problems: Section 1.1: 1, 6, 13, 17, and 21 Section 1.2: 2, 3, 9, 11, and 20 Section 1.3: 3, 5, 8, 9, 12, 15, and 17 Section 1.4: 5, 9, 11, 16, 19, and 22 Section 1.5: 1, 5, 9, 13, 19, and 26 Section 1.6: 1 and 11
2	Linear Independence and Transformations	Lab exercises, lectures, and facilitated discussions	Chapter 1: Sections 7 to 10	Solve the following problems: Section 1.7: 1, 4, 7, 10, 13, 24, and 33 Section 1.8: 2, 5, 10, 13, 17, 23, and 31 Section 1.9: 3, 6, 11, 15, and 19 Section 1.10: 1, 7, and 13
3	Matrix Algebra and Applications	Lab exercises, lectures, and facilitated discussions	Chapter 2: Sections 1 to 3 and 8	Solve the following problems: Section 2.1: 1, 4, 7, 8, 12, and 28 Section 2.2:3, 7, 14, 19, 21, and 33 Section 2.3: 2, 5, 17, 26, 33, and 37 Section 2,8: 1, 6, 11, 13, and 23
4	Determinants	Lab exercises, lectures, and facilitated discussions	Chapter 3: Sections 1 to 3	Solve the following problems: Section 3.1: 1, 10, 16, 24, and 40 Section 3.2: 7, 15, 21, 25, and 31 Section 3.3: 2, 5, 10, 20, and 24
5	Vector Spaces and Subspaces (Part 1)	Lab exercises, lectures, and facilitated discussions	Chapter 4 Section 1	Solve the following problems: Section 4.1: 1, 4, 9, 12, 15, 21, 23, 26, 32, and 35
6	Vector Spaces and	Lab exercises,	Chapter 4	Solve the following problems:

	Subspaces (Part 2)	lectures, and facilitated discussions	Sections 2 to 5	Section 4.2: 1, 4, 5, 11,, 17, 23, and 29 Section 4.3: 3, 7, 11, 13, and 19 Section 4.4: 1, 5, 10, 13, and 27 Section 4.5: 2, 7, 9, 11, 15, and 21
7	Eigen values and Eigenvectors	Lab exercises, lectures, and facilitated discussions	Chapter 5 Sections 1 to 3	Solve the following problems: Section 5.1: 1, 3, 5, 13, 17, 23, and 26 Section 5.2: 3, 12, 19, 20, and 21 Section 5.3: 1, 3, 5, 13, 19, and 27
8	Orthogonality	Lab exercises, lectures, and facilitated discussions	Chapter 6 Sections 1 to 3	Solve the following problems: Section 6.1: 1, 3, 9, 13, 17, 23, and 27 Section 6.2: 1, 5, 7, 13, 15, and 29 Section 6.3: 1, 3, 5, 13, 16, 19, and 21

Student Evaluation Grid

Assignment	Value (percent of overall course grade)
First Week Assignment	5%
Weekly Problem Sets	35%
Midterm Exam	20%
Final Exam	25%
Participation	15%
TOTAL	**100 %**

Regis University Grading Scale

A	93 – 100	C	73 - 76
A-	90 – 92	C-	70 – 72
B+	87 – 89	D+	67 – 69
B	83 – 86	D	63 – 66
B-	80 – 82	D-	60 – 62
C+	77 – 79	F	59 or Lower

Attendance/Participation

Facilitator will provide specific attendance/participation policies in accordance with the official attendance policy of the School of Computer and Information Sciences.

Attendance Policy of the School of Computer and Information Sciences
Student participation is essential. Students are responsible for all material covered in each week. Students' interaction is central to the learning experience and imparts critical observations and knowledge to all concerned.

If a student misses more than one week of class participation without prior faculty approval, the facilitator may assign a failing grade.

Late Assignments/Assignment Revisions
Facilitator will provide specific late assignments/assignment revisions policies not to exceed the maximum official policy of the School of Computer and Information Sciences.

Late Assignments/Assignment Revisions Policy of the School of Computer and Information Sciences
Occasionally, work, health, and family obligations have priority over course work. If you experience these issues, please notify the facilitator in advance. If an assignment is due, please make every effort to submit your assignment on time. In the case of a negotiated absence or delay, assignments turned in by the agreed upon due date will receive full credit.

If you have not negotiated with the facilitator, assignments turned in late will be graded, and then reduced by 25%. Assignments will not be accepted later than one week after due date.

Incomplete Grade
An incomplete grade is given at the discretion of the facilitator. A facilitator is not required to assign an incomplete grade. Each occurrence will be negotiated on an individual basis.

Incomplete Grade Policy of the College of Professional Studies
A grade of Incomplete or 'I" denotes that the required work for the course is incomplete due to unforeseen circumstances. Unforeseen circumstances mean, for example, that an accident, an illness, a death, or a major life transition has occurred. This grade is awarded at the discretion of the instructor and is submitted with an alternate grade of "I/F."

A student must submit a written request to the instructor asking that an incomplete grade be assigned prior to the end of the term in which the course is taken. The course instructor will determine the length of time for course completion, and the remaining requirements to complete the course; however the maximum time for completion is the end of the following semester.

Faculty will notify students in writing of the approval/denial of their request and provide instructions and guidelines required to resolve the incomplete grade. If the course work is not completed by the end of the next applicable semester and a grade change form submitted by the instructor, the incomplete grade reverts to the alternate grade assigned by the instructor and is calculated in the grade point average.

Tutoring Information:

Occasionally students need additional assistance with course content. Tutorial assistance is available to students through SmartThinking, writing assistance and personal tutoring. SmartThinking provides every student with ten hours of free online tutoring in writing, math, statistics, economics and accounting each year. Writing assistance is available in the form of Roving Writing Tutors and a variety of writing workshops. Individual personal tutors are available in a variety of discipline areas with fees and arrangements made between the individual student and tutor. For access to these services, go www.regis.edu , Current Student, Services for CPS Students, Academic Tools, Tutoring/SmartThinking.

Disability Statement:
Equal access to education means equal opportunity to learn in an academic setting. Under Section 504 of the 1973 Rehabilitation Act and the Americans with Disabilities Act of 1990, students with documented disabilities who are otherwise qualified have a right to equal access to classes at Regis University. If you have a documented disability requiring academic adjustments for this class, please contact the Director of the Office of Disability Services at **303-458-4941** for discussion of disability issues. Since accommodations are not retroactive, it is suggested that you contact the Disability Services Office as soon as possible; adequate lead-time may be required to receive accommodations, including review of appropriate documentation.

Academic Dishonesty
Regis University is committed to intellectual integrity in its academic pursuits. Sanctions may, therefore, be imposed by the College for Professional Studies faculty or schools for cheating (defined as using inappropriate sources of information on a test) or plagiarism (defined as presenting as one's own, the ideas, words, or products of another). Such sanctions may include a failing grade on the assignment, failure of the course, or expulsion of the student from the course or the department.

Non-Public, Confidential or Proprietary Information Compliance
For class projects that relate to a student's employment, the student is responsible for complying with all applicable policies of the employer, including those related to non-public, confidential or proprietary information